

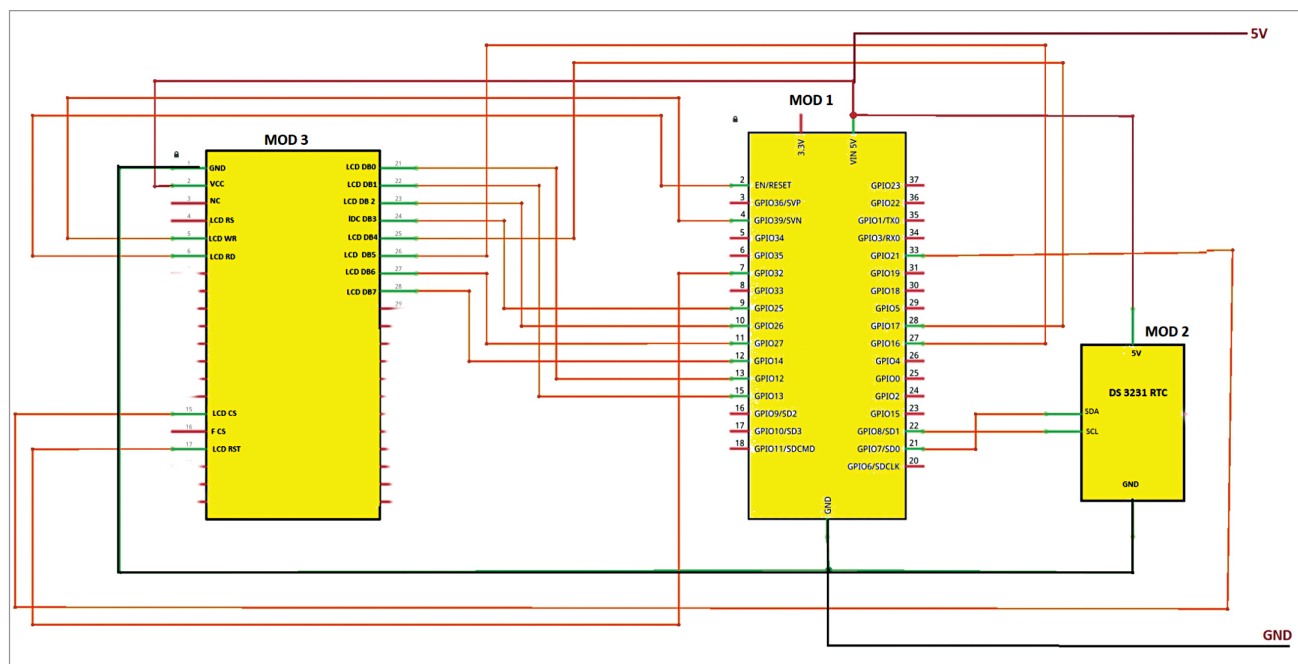


Fig. 4: Circuit diagram



Fig. 5: Author's prototype

easily on Arduino Uno, a millisecond display is not feasible on Arduino Uno as it lacks computing power. So, the ESP32 and DS RTC module are used to help the device in calculating down to milliseconds and display it in real time.

So, unlike other smart clocks you may have in your home, this clock will help you get an accurate time up to milliseconds. You can use it as a stopwatch as well.

The author's prototype is shown

in Fig. 1, while Fig. 2 shows the TFT display used in the project. The components required to build it are mentioned under the Bill of Material table.

Software

First the code is made to communicate with RTC DS3231 module and fetch the time from it. After that communication is established with the LCD display. The loop makes the clock function, and it displays the time accurately to a millisecond.

After preparing the source code, upload it to ESP32 by selecting the ESP32S board and ESP port number in Arduino IDE. Fig. 3 shows a snippet of the source code.

After uploading the source code, connect the RTC DS3231 module to I2C pins of ESP32 board. Also connect the TFT LCD display with the

ESP32 board. The pinouts of TFT and ESP32 board are shown under Table 2.

Circuit and testing

Fig. 4 shows the circuit diagram of the project, which is built around ESP32S (MOD1), ILI9488 TFT display (MOD3), and RTC DS3231 module (MOD2).

The clock can be powered with 5V battery or a USB. After powering up, the time is calculated by RTC DS3231. This RTC has the capability of measuring temperature as well, which is also displayed on the TFT.

The time duration between consecutive seconds is divided by 1,000 to calculate the time in milliseconds and displayed on the TFT. The simple line drawing is tricked to create a thick line, which is put into variable function. Fig. 5 shows the final TFT clock made by the author. **EFY**

EFY Note

The source code of this project is available for free download at www.electronicsforu.com



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